Simon Gardier

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Education

University of Liège — Liège, Belgium
M.Sc. in Computer Science (Intelligent Systems) — *Sep 2024 – Jun 2026*

- Thesis: Feed-Forward Novel View Synthesis for Soccer Scenes with Priors, **17/20** (magna cum laude).
- **Award**: AIM Award for the Best Master's Thesis in Computer Science.
- **Scholarship**: Alexandre Ketin's Scholarship
- Grade average: 15.38/20 (cum laude)
- Relevant coursework: Computer Vision, Deep Learning, Intelligent Robotics, Machine Learning Systems Design, Distributed Systems, Information Theory, Compilers, OOP for Mobile Applications.

University of Liège — Liège, Belgium
B.Sc. in Computer Science — *Sep 2020 – Sep 2024*

- Grade average: 13.45/20
- Relevant coursework: Operating Systems, Computation Structures, Embedded Systems, Parallel Programming, DSA, Databases, Networking, Graph Theory, Discrete Mathematics, Statistics, Probability, Physics.

Research Experience

EVS Broadcast Equipment — Liège, Belgium
Innovation Intern — *Dec 2025 – Jun 2026*

- Designed and trained a modular feed-forward 3D reconstruction pipeline for soccer broadcast scenes, with novel-view synthesis running in **~1 s per scene, down from several minutes, on a single GPU.**
- Implemented pitch grounding without ground-truth, allowing operators to work in 3D world coordinates.
- Reviewed the state of the art and benchmarked **4 models** on downstream tasks using a custom evaluation pipeline.

Toyota Motor Europe — Brussels, Belgium
AI Research Intern — *Sep 2025 – Dec 2025*

- Proposed and implemented dynamic token selection for vision-language model personalization, improving recall by **up to 30% for challenging subjects**.
- Conducted **100+ ablation experiments** with PyTorch, InternVL3, and Transformers.

Publications

P1 **Ego: Embedding-Guided Personalization of Vision-Language Models**
Soroush Seifi, Simon Gardier, Vaggelis Dorovatas, Daniel Olmeda Reino, Rahaf Aljundi
*Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026*

Teaching Experience

University of Liège — Liège, Belgium
Teaching Assistant (First Programming Project) — *Sep 2021 – Sep 2024*

- Taught practical Python sessions for approximately **20 students**, helping them debug code and develop problem-solving skills.

Volunteering

RoboCup Montefiore Team, University of Liège — Liège, Belgium
Computer Scientist — *Sep 2023 – Dec 2025*

- Implemented a Cortex-M4 bootloader in C with libopencm3, supporting firmware updates over CAN bus.
- Built a sim-to-real pipeline that controls a robot from a Blender virtual clone; the robot performed in a short film screened to **2,000+ viewers** as part of an engineering-student theatre interval.

Professional Services

Technical Reviewer - *Build a CPU from Scratch* by Amane Uehara, Manning Publications (forthcoming) — *Jul 2026*

Selected Projects

Monte Maps - Indoor Visual Localization — *Jun 2026*

- Built a camera localization pipeline combining MegaLoc image retrieval with a COLMAP point cloud using Python and PyTorch, allowing localization in **real-time on a personal laptop**.

YOLOv5 (m) from scratch — *Jun 2025*

- Implemented the backbone and neck of the YOLOv5 model from scratch, applying ML and deep learning concepts. Trained the model on a dataset of 3k Minecraft images for ore detection.

Chess Video Analysis — *Dec 2024*

- Combined computer vision and machine learning to reconstruct chess game states from fixed and moving videos using Python, OpenCV, and scikit-learn.
- Achieved the fastest execution time in a competition involving ~40 students.

Dance Dance Revolution in Assembly — *Jun 2024*

- Built a rhythm game in **~7,000 lines of PIC assembly** on a **32 MHz microcontroller with 2 KB RAM**.
- Engineered a custom-built dance-pad hardware with the game software; awarded **20/20** (seldom given).

Other

System & Tools: Git, LaTeX, Linux, Blender, libopencm3.
Extracurricular activities: Fire cadet school, Scout leader, Bouldering, Rowing.
Languages: French (fluent), English (fluent), Dutch (basic)